SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

ANUPALAN – AI Compliance & Governance Command Centre

- **Problem Statement ID – SIH26024**
- **Problem Statement Title-** AI-Based Smart Governance and Compliance Monitoring System for Coal Mines
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** ◄ **ADD TEAM ID FROM SIH PORTAL**
- **Team Name (Registered on portal)- NeuraForge**

Coal India Limited · Ministry of Coal | Presidency University, Bengaluru

ANUPALAN – AI Compliance Command Centre

NeuraForge

SMART INDIA HACKATHON 2026

Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- **Digital Rulebook.** Regulation-as-Code parses the Mines Act 1952, CMR 2017, DGMS circulars and EC/CTO conditions into tracked duties with owner, deadline and required evidence.
- **Offline-first field evidence.** Geo-tagged, timestamped, hash-chained capture that works underground and syncs on reconnect.
- **Predictive risk engine.** Scores which mine and which clause is most likely to fail next, from inspection and violation history.
- **Hazard early-warning, no new hardware.** Reads CIL's existing ETMS, methane detectors, strata monitors and geo-fenced vehicle telemetry to flag inundation, roof and ground movement, gas build-up, seismic shift and equipment failure as live breaches.
- **Geospatial compliance layer.** Lease, forest and eco-sensitive polygons in PostGIS; satellite change detection catches excavation outside sanctioned limits and screens expansion zones before they are planned.

How it addresses the problem

- **Fragmentation → one source of truth.** Safety, environment, production and labour duties in a single auditable system instead of spreadsheets and paper registers.
- **Reactive → predictive.** Hazard telemetry and inspection history combine into a score that flags the breach before it is recorded.
- **Weeks → hours.** The inspection-to-report cycle collapses, and every entry traces back to the exact statutory clause.
- **Self-declared → verifiable.** Tamper-evident evidence stands up in a DGMS enquiry; boundary breaches are caught from imagery, not complaints.
- **Scarce inspectors, better aimed.** Risk ranking tells DGMS which mine to visit first instead of inspecting round-robin.

How ANUPALAN Works

Statute to field action, as one closed loop

1 **Ingest the law**
Mines Act 1952, CMR 2017, DGMS circulars, EC & CTO conditions parsed by NLP

▼

2 **Auto-build obligation ledger**
Every clause becomes a tracked duty with owner, deadline and required evidence

▼

3 **Capture evidence in the field**
Offline-first mobile app; geo-tagged, timestamped, hash-chained proof

▼

LIVE HAZARD SIGNALS — no new hardware
Read from instruments CIL already runs: ETMS, local methane detectors, strata monitors, geo-fenced vehicle telemetry
Flooding | Water ingress | Roof collapse | Ground instability | Seismic | Toxic gas | Equipment failure

▼ feeds ▼

4 **Score risk before breach**
Predicts which mine fails which clause next; vision triage filters false reports

▼

5 **Alert, escalate, auto-file**
Officer-approved statutory returns to DGMS, MoEFCC and CPCB

↻ **Continuous loop.** Every closed action retrains the risk model and updates the live compliance posture.

Innovation and uniqueness of the solution

- **A rulebook that updates itself.** The Mines Act 1952 is repealed the day the OSH&WC Code 2020 is notified. Hard-coded rules break; a parsed corpus re-derives every duty overnight.
- **Predicts the breach, does not record it.** Risk scoring fed by hazard telemetry CIL already collects, so it needs no new sensors.
- **Evidence that survives an enquiry.** Offline, geo-tagged, hash-chained and pre-screened by vision.

NeuraForge

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

Technologies to be used (e.g. programming languages, frameworks, hardware)

Frontend
React 18 + TypeScript, Tailwind CSS, Recharts / D3, Leaflet maps

Mobile
React Native, SQLite / WatermelonDB offline-first, geo-tagged camera

Backend
FastAPI (Python), Node.js services, REST + GraphQL, MQTT / Kafka ingest

AI / ML
pgvector RAG + sentence-transformers, XGBoost risk model, YOLOv8 vision, time-series hazard detection

Data
PostgreSQL + PostGIS, TimescaleDB, Redis, MinIO object store

Infra & Security
Docker, Kubernetes, NIC MeghRaj cloud, Parichay SSO, RBAC, AES-256, hash-chained audit log

Methodology and process for implementation (Flow Charts/Images/ working prototype)

DATA SOURCES
- **Statutory Corpus** — Mines Act 1952 · Coal Mines Regulations 2017 · DGMS circulars
- **EC & CTO Conditions** — PARIVESH · CPCB consent conditions, forest clearance
- **Field Inspections** — Geo-tagged photo & form capture, offline-first
- **Mine Safety Instruments** (NEW) — ETMS · local methane detectors · strata monitors · geo-fenced vehicle telemetry
- **Satellite & GIS** (NEW) — Lease boundaries · forest and eco-sensitive zone layers · change detection imagery
- **Legacy Systems** — CIL CSIS · CMSMS · Safety Report Portal

INGESTION LAYER
- **API Gateway** — REST / GraphQL, rate-limited
- **Offline Sync Queue** — Conflict resolution on reconnect
- **Telemetry Stream** — MQTT / Kafka sensor ingest
- **ETL & Document Parser** — PDF / OCR / circular normaliser

AI / ML
AI CORE — THE DIFFERENTIATOR
- **Regulation-as-Code** — RAG over the statute corpus; clause → duty, owner, due date, evidence type
- **Compliance Risk Engine** — XGBoost on inspection and violation history → predicted failure score per mine
- **Hazard Early-Warning** (NEW) — Time-series on inundation, strata movement, gas, seismic and equipment health
- **Geo-Compliance** (NEW) — Boundary-breach detection and clearance-aware zone screening in PostGIS
- **Vision Triage** — YOLOv8 PPE, hazard and site checks; cuts the false-positive report load

DATA LAYER
- **PostgreSQL + PostGIS** — Lease, spatial & mine data
- **TimescaleDB** — Sensor time-series history
- **pgvector Store** — Clause embeddings for RAG
- **MinIO Object Store** — Evidence photos, returns
- **Hash-Chained Audit Log** — Tamper-evident, enquiry-ready

DELIVERY & ACTION
- **Mine & Area Dashboard** — Live obligation ledger
- **Corporate View** — CIL + 8 subsidiaries
- **Regulator Portal** — DGMS · MoEFCC · CPCB
- **Field Mobile App** — Offline capture & alerts
- **Auto Statutory Returns** — Officer-approved filing

NeuraForge

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Analysis of the feasibility of the idea

- **No new hardware.** The sensors already exist. CIL runs ETMS, methane detectors, strata monitoring and geo-fenced vehicle tracking today; we read them.
- **Zero licence cost.** Entire stack is open source and deploys on existing NIC MeghRaj or CIL data-centre capacity.
- **Inputs are already public.** Statutes, DGMS circulars, EC and CTO conditions are published on DGMS, MoEFCC and PARIVESH.
- **No retraining needed.** Screens mirror the Form A/B registers field staff already fill; regional language support.
- **Buildable in 36 hours.** Working prototype on a seeded corpus of the top 50 statutory duties and one demo mine.
- **Proven components.** Nothing is research-grade — RAG, gradient boosting and YOLOv8 are production-standard.
- **Phased rollout.** Pilot one subsidiary, then scale. No big-bang migration across 313 mines.

Potential challenges and risks

- **No connectivity underground.** Field capture cannot assume a network.
- **Legal text is messy.** Long, unstructured and amended often by fresh circulars.
- **AI misreading the law.** A hallucinated statutory interpretation is a serious liability.
- **Sensor data is patchy.** Instruments differ by mine, by vintage and by vendor; some feeds are analogue.
- **Adoption resistance.** Field staff are used to paper and may distrust a scoring system.
- **Data sensitivity.** Safety and violation data spans eight subsidiaries with different owners.
- **Thin training history.** Early risk scores have limited labelled violation data to learn from.

Strategies for overcoming these challenges

- **Offline-first architecture.** Local queue with conflict resolution; uploads on reconnect.
- **Versioned corpus, scheduled re-index.** A new circular automatically recomputes the duties it touches.
- **Human-in-the-loop, always.** AI drafts and cites the clause; a qualified officer signs. It never auto-files a return.
- **Adapter pattern for instruments.** One driver per sensor type, with manual entry as the fallback so no mine is blocked.
- **Mirror the familiar.** Existing register formats, 15-minute onboarding, no workflow change.
- **Isolate by design.** On-premise deployment, RBAC, subsidiary-level partitioning, AES-256 at rest.
- **Rules first, model later.** Ship deterministic rule checks on day one; the model augments as data accumulates.

36-Hour Grand Finale Build Plan — 6 members, 4 parallel tracks, integration checkpoint every 6 hours

HOUR 0–6 Foundation	HOUR 6–18 Core Build	HOUR 18–30 AI Integration	HOUR 30–36 Demo & Hardening
Schema, auth, RBAC. Seed corpus of the top 50 statutory duties loaded and embedded. Demo mine seeded with lease polygon.	Obligation ledger and dashboards. Offline-first mobile capture with geo-tagging. Sensor feed simulator streaming live.	RAG clause mapping, XGBoost risk score, hazard early-warning, PostGIS boundary breach, YOLOv8 triage wired to live data.	Auto-drafted returns, audit trail, three scripted judge scenarios, fallback recording, dry-run of the pitch.

NeuraForge

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

313	8	3.3 lakh	781 MT	16
CIL working mines (April 2024)	producing subsidiaries on one platform	workforce, incl. 1.15 lakh on contract	CIL output, FY 2024-25	CIL units run without valid clearance (CAG)

Potential impact on the target audience

- **Mine managers & safety officers** — every duty, deadline and piece of evidence in one place, with hazard alerts arriving before an inspection does.
- **CIL corporate & its subsidiaries** — live compliance posture across 313 mines and early warning ahead of a recorded violation.
- **DGMS, MoEFCC & CPCB** — verified, timestamped, clause-linked evidence in place of self-declared PDFs, so scarce inspector time goes to the highest-risk mines.
- **3.3 lakh workers, contractors included** — the safety committee's own minutes record contract labour being left out of safety committees; here every duty carries a named owner.
- **Citizens near coalfields** — clearance conditions and lease boundaries monitored continuously instead of self-certified twice a year.

Benefits of the solution (social, economic, environmental, etc.)

- **Social** — preventive action against the 25 fatal accidents recorded at CIL mines in 2025, and safer conditions for a 3.3 lakh workforce.
- **Economic** — fewer stoppages, show-cause notices and penalties; sharply lower paperwork and audit-preparation cost.
- **Environmental** — EC and CTO conditions monitored continuously: air and water quality, land reclamation, plantation targets.
- **Governance** — a complete, tamper-evident audit trail makes accountability provable rather than asserted.
- **Scalable** — the same engine covers metalliferous and non-coal mines under DGMS, and any sector governed by statutory conditions.
- **SDG alignment** — SDG 8 (decent work and safe workplaces), SDG 9, SDG 12, SDG 13 and SDG 16 (accountable institutions).

Measurable targets for a one-subsidiary pilot

Inspection to filed report	False-positive field reports	Statutory returns auto-drafted	Mines with a live risk score
weeks → under 24 hours	54% → under 15%	0% → 80%	0 → all 313

NeuraForge

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

Details / Links of the reference and research work

1 CAG of India — Report No. 12 of 2019, environmental impact of mining in CIL

The auditor found 16 CIL units operating without valid statutory clearance, 6 of 7 subsidiaries with no environment policy, and only 58 of 96 required air-quality stations installed.

cag.gov.in — Report No. 12 of 2019

2 Standing Committee on Safety in Coal Mines — 49th meeting minutes, Feb 2025

80 fatal accidents occurred between sittings, and the committee recorded dust-suppression discrepancies between documented records and field conditions. That gap is the problem we close.

coal.gov.in/sites/default/files/2025-02/10-02-2025b-wn.pdf

3 Ministry of Coal — Annual Report, Ch. 14 "Safety in Coal Mines"

CIL fatality rates, the National Coal Mines Safety Report Portal, CSIS, and the instrumentation ANUPALAN reads: ETMS, methane detectors, strata monitoring, geo-fenced vehicle tracking.

coal.gov.in/sites/default/files/2025-02/chap14AnnualReport2025en2.pdf

4 The Mines Act 1952 · Coal Mines Regulations 2017 · Mines Rules 1955 · OSH&WC Code 2020

The statutory corpus mapped clause-by-clause into the obligation ledger. The Code repeals the Mines Act on notification, which is why the rulebook is parsed rather than hard-coded.

indiacode.nic.in

5 PIB — Coal Mine Surveillance & Management System and Khanan Prahari

Of 462 reports, 251 were verified false and 77 genuine; a later release records only 78 of 483 complaints acted upon. That verification load is what our triage targets.

pib.gov.in/PressReleasePage.aspx?PRID=1898772

6 Directorate General of Mines Safety, Ministry of Labour & Employment

The enforcement body: 8 zones, 38 regional offices, roughly 19,012 registered mines. Source of the circulars the Digital Rulebook ingests.

dgms.gov.in

7 PARIVESH and CPCB — clearance and consent conditions

Environmental clearance and Consent to Operate conditions, plus the half-yearly compliance report that is filed today as a self-declaration.

parivesh.nic.in · cpcb.nic.in

8 Ministry of Coal Annual Report Ch. 1 and 8 — scale of the estate

313 CIL working mines (131 underground, 168 opencast, 14 mixed), 8 producing subsidiaries, 781 MT in FY 2024-25, 2.14 lakh employees and 1.15 lakh contract workers.

coal.gov.in — Annual Report, Chapters 1 and 8